

1. BUY & CHECKOUT PROCESS (ENGINEERING FOCUS)

1.1 Core Business Rules

Payment Rules

- All orders require:
 - **50% deposit at checkout** (*mandatory*)
 - OR **Afterpay (Buy Now, Pay Later)** option
- Remaining **50% balance is collected at pickup**
 - Accepted methods:
 - Cash
 - Credit/Debit Card (via POS)

Pre-Order Definition

- A **pre-order = Any order where a deposit is collected prior to fulfillment**
 - Inventory is **reserved but not fully deducted** until fulfillment
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1.2 Checkout Flow (End-to-End)

Step-by-Step Process

1. Browse Products

- Filter by category (Fish, Shellfish, Combo Boxes)
- View availability, price per lb/unit, and pickup options

2. Add to Cart

- Quantity selection
- Dynamic pricing (weight-based or fixed pricing)

3. Cart Validation

- Validate:
 - Product availability
 - Pickup eligibility
 - Supplier constraints

4. Select Pickup Details (MANDATORY)

- Location
- Date
- Time slot

5. Checkout Page

- Customer Info:
 - Name
 - Phone
 - Email
- Order Summary
- Payment Options:
 - **50% Deposit (default)**
 - **Afterpay**

6. Payment Processing

- Calculate:

$\text{deposit_amount} = \text{total_order} * 0.5$

- Generate:
 - Order ID
 - Payment transaction ID

7. Order Confirmation

- Status: PRE_ORDER_CONFIRMED
 - Notifications:
 - Email/SMS confirmation
 - Pickup instructions
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END-TO-END SYSTEM FLOW (SIMPLIFIED)

```
graph TD; A[Customer Browsing] --> B[Add to Cart]; B --> C[Select Pickup (Location/Date/Time)]; C --> D[Checkout (50% Deposit / Afterpay)]; D --> E[Order Created (PRE_ORDER_CONFIRMED)]; E --> F[Inventory Reserved]; F --> G[Supplier Fulfillment]; G --> H[Order Ready]; H --> I[Customer Pickup + Final Payment]; I --> J[Order Completed];
```

1.3 Order State Machine (Backend Logic)

Status	Description
CART	Items added, not checked out
PENDING_PAYMENT	Awaiting deposit

Status	Description
PRE_ORDER_CONFIRMED	Deposit received
READY_FOR_PICKUP	Order fulfilled and ready
COMPLETED	Paid in full at pickup
CANCELLED	Order voided

1.4 Payment Scenarios

Scenario A: Standard Deposit

- Total = \$200
- Deposit = \$100 (online)
- Balance = \$100 (paid at pickup)

Scenario B: Afterpay

- Full amount financed
- Marketplace receives payment upfront (via provider)
- Order status = PAID_FULL

Scenario C: No Pickup Payment

- If balance unpaid:
 - Status → ON_HOLD
 - Configurable grace period (e.g., 24 hours)
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1.5 Edge Cases

- **✗ No pickup selected → Block checkout**
- **✗ Inventory unavailable → Prevent purchase**

-  **Partial fulfillment** → **Notify + adjust balance**
 -  **Cancellation rules:**
 - **Refund deposit (configurable policy)**
 - **Or store credit**
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2. ♀ **PICKUP LOCATIONS (ENGINEERING + OPERATIONS)**

2.1 Customer-Facing Rules

- **Customers MUST select:**
 - **Pickup Location**
 - **Date**
 - **Time Slot**
 - **Only valid combinations should be selectable**
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2.2 Admin Configuration (Owner Panel)

Pickup Location Object

```
{  
  "location_id": "LOC001",  
  "name": "Downtown Dock Pickup",  
  "address": "123 Harbor St",  
  "operating_days": ["Friday", "Saturday"],  
  "time_slots": ["10:00 AM - 12:00 PM", "12:00 PM - 2:00 PM"],  
  "capacity_limit": 50  
}
```

2.3 Business Rules

- **Each location must define:**
 - **Operating days**
 - **Time slots**
 - **Order capacity per slot**
 - **Orders must:**
 - **Not exceed slot capacity**
 - **Align with product availability**
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2.4 Process Flow

Customer Flow

- 1. Select location**
- 2. Select date (filtered by availability)**
- 3. Select time slot**

System Validation

- **Check:**
 - **Slot capacity**
 - **Inventory readiness**
 - **Supplier delivery alignment**
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2.5 Admin Capabilities

- **Create/Edit/Delete pickup locations**
- **Set blackout dates**
- **Adjust capacity limits**
- **View orders per location/time slot**

3. CUSTOMER EXPERIENCE (CX) & PRODUCT MANAGEMENT (OWNER FOCUS)

3.1 What Defines Great CX in Seafood Marketplace

Transparency & Trust

- Show:
 - Source of seafood (wild-caught vs farm-raised)
 - Catch date / freshness indicator
 - Supplier details

Clear Pricing

- Per lb vs per item clarity
- Estimated vs final weight explanation

Real-Time Communication

- Order confirmations
- Pickup reminders (24 hrs + same day)
- Delay notifications

3.2 Product Management Best Practices

Product Structure

Each product should include:

- Name
- Category
- Price (per lb or unit)
- Minimum order qty

- Images
 - Description (taste, cooking methods)
 - Source (region, supplier)
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3.3 Inventory Management Rules

- Inventory Types:
 - Pre-order inventory (soft allocation)
 - Physical inventory (hard allocation)
 - Deduction Logic:
 - At order → reserve inventory
 - At fulfillment → deduct inventory
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3.4 Supplier Management

Supplier Data Model

- Supplier Name
- Contact Info
- Product Supply List
- Delivery Schedule
- Compliance Certifications

Business Rules

- Suppliers must:
 - Meet FDA/USDA seafood handling standards

- Provide traceability records
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3.6 Pickup Experience (CX)

- Fast, organized pickup flow:
 - Order lookup (QR code / phone number)
 - Separate lines:
 - Paid in full
 - Balance due
 - On-site POS for balance payments
 - Packaging:
 - Insulated bags
 - Ice packs
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3.7 User Management (Admin Panel)

Roles

Role	Permissions
Admin	Full access
Manager	Orders, inventory, suppliers
Staff	Pickup & fulfillment only

3.8 Order Management (Owner Workflow)

Daily Flow

- 1. Review incoming pre-orders**
 - 2. Confirm supplier availability**
 - 3. Allocate inventory**
 - 4. Prepare fulfillment list**
 - 5. Mark orders as READY_FOR_PICKUP**
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3.9 Reporting & Metrics

Track:

- Total Sales**
- Pre-orders vs Completed Orders**
- Pickup location performance**
- Inventory turnover**
- Customer retention**